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1

Introduction

Qualitative phase imaging (QPI) techniques, such as phase-contrast (PhC) microscopy [1] and differential interference contrast (DIC) microscopy [2], enable the visualization of fine structures in unlabeled biological transparent tissues that are difficult to observe using conventional bright-field microscopy by converting small phase variations into detectable intensity changes [3] and can be performed at lower illumination powers compared to dark-field microscopy [4]. In contrast, QPI techniques directly measure intrinsic optical properties of matter, enabling label-free characterization and spectroscopic analysis, with applications ranging from live-cell studies and tissue identification (both *ex vivo* and *in vitro*) to rapid pathological assessment [5]. Recently quantitative differential phase contrast (qDPC) microscopy has gained great attention as a non-interferometric technique that employs partially coherent illumination to achieve lateral resolution up to twice the coherent diffraction limit [6], and incorporates multi-angle illumination to improve axial sectioning relevant in 3D implementations of tomographic reconstruction [7]. To extend DPC imaging to thick biological tissues, quantitative oblique back-illumination microscopy (qOBM) modifies conventional DPC by treating the specimen as a stack of thin slices, each satisfying the weak object approximation (WOA) [8]; the corresponding phase optical transfer function (OTF) can be approximated using the transport of intensity (TIE) theory by ignoring multiple scattering effects [9–11].

The thesis focuses on 2D deconvolution-based phase reconstruction in the epi-direction, implemented primarily via the Wiener filter or Hilbert transform, both previously applied within the DIC framework [2]. Accurate phase quantification via this approach provides access to refractive-index variations, cellular morphology, and dry mass [12]. qOBM has demonstrated biomedical potential in applications ranging from virtual histological staining of tissue biopsies [13] and volumetric skin imaging [14] to real-time diagnosis of brain tumors [15].

The remainder of this thesis is organized as follows: Chapter 2 examines the characteristics of DPC images experimentally acquired using a scanning-laser qOBM microscope across various biological and technical specimens, followed by the development of a numerical simulation model for the DPC microscope. Chapter 3 investigates two distinct implementations of the deconvolution algorithm for phase map retrieval: a Hilbert-transform approach that leverages the signum function for phase decoding, and a Wiener-filter approach that utilizes the optical transfer function (OTF) derived from the simulated DPC microscope. Finally, Chapter 4 provides a comparative analysis of the phase reconstructions obtained from these two deconvolution methodologies.

2

DPC image generation and material selections

Differential phase contrast (DPC) microscopy utilizes asymmetric oblique illumination to convert invisible phase shifts into visible intensity variations [16]. By subtracting intensity images captured under opposing illumination angles, the system generates a raw DPC image whose contrast is directly proportional to the phase gradient of the specimen. Under the Weak Object Approximation (WOA)—where the complex transmittance is given by $t(x) \approx 1 + i \cdot \phi$ [8]— The subtraction operation enables a linear description of the DPC microscope. In addition to linearizing the system response, the differential subtraction cancels out background noise, illumination non-uniformities, out-of-focus details, and optical aberrations [6, 17]. The resulting output clearly outlines specimen boundaries and edges. This imaging modality is implemented using an inverted scanning microscope configured for quantitative phase imaging in reflection mode, as shown in Figure 2.1.

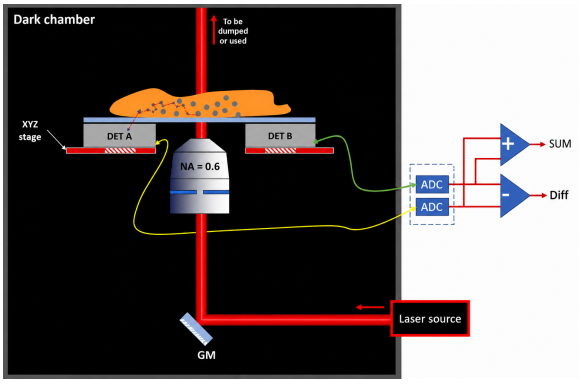
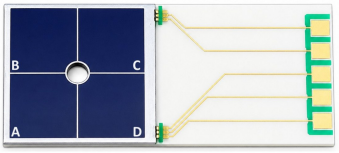


Figure 2.1: Schematic of the sqOBM microscope layout.

A continuous-wave Gaussian laser beam at a wavelength of 1030 nm was directed upward and raster-scanned across the sample via galvanometer mirrors before being focused by the microscope objective (Nikon 50x NA = 0.8) into the tissue, with the detection path easily reconfigurable for forward-transmission operation by repositioning the detector on the opposite side of the sample. For epi-detection, the four-quadrant detector (4QD) module was affixed to the underside of the glass slide supporting the specimen using adhesive tape, allowing it to measure the center-of-mass displacement of the backscattered light distribution; the entire assembly was enclosed in a dark chamber to minimize stray light and enhance detec-

tion sensitivity. Mounted on an XYZ translation stage, the setup enabled precise alignment and flexible multifocal-plane imaging, facilitating systematic volumetric acquisition of phase-gradient data across different sample depths to perform.

The 4QD exhibits wideband spectral sensitivity that is maximized in the near-infrared, with a peak responsivity of approximately 430 mA/W at 850 nm [18]. The total active area of the photodiode is 100 mm^2, with each of the four quadrants measuring 4 mm $\times$ 4 mm and mounted on a ceramic support. Each quadrant operates as an independent photodetector and measures the intensity of backscattered light reaching the detector from the sample. A circular hole with a diameter of 2.3 mm is mechanically drilled through the center of the detector to allow the laser beam to pass through. At each scanning position, the photocurrent generated by each quadrant is converted into a voltage using a transimpedance amplifier. The four measured signals are used to calculate the orthogonal differential phase gradients (I_{DPCx}, I_{DPCx}) and the bright field signal (I_{SUM}), as follows:



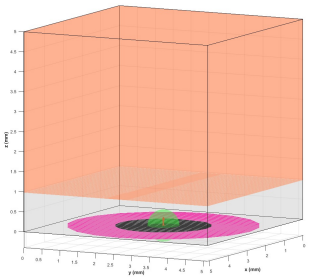
$$\begin{cases} I_{DPCx} = (I_A + I_B) - (I_C + I_D), \\ I_{DPCy} = (I_B + I_C) - (I_A + I_D), \\ I_{SUM} = I_A + I_B + I_C + I_D \end{cases} \tag{1}$$

where I_A, I_B, I_C, and I_D denote the photocurrents measured by the respective quadrants (A, B,C, D as depicted in), corresponding to the backscattered light reaching the detector under sequential asymmetric illumination. These computations are implemented in a MATLAB script, with the resulting output represented as a grayscale intensity value ranging from 0 to 255. A value of 0 corresponds to no detected light or equal intensity across all quadrants, whereas a value of 255 indicates that the detection is biased toward a given quadrant. The pixel resolution of the resulting image is determined by the number of readouts, for example, 120, 512, or 1024 pixels, acquired during each laser-scanning cycle.

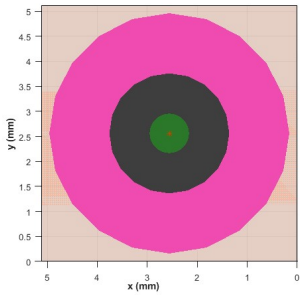
2.1 Simulation model of the qOBM

The computational model consists of a $256 \times 256 \times 250$ voxel domain corresponding to a physical volume of $5.12 \times 5.12 \times 5.0$ mm^3, with a voxel size of 20 μm, as shown in Figure 2.2 (a-c). To approximate a semi-infinite medium, perfectly absorbing boundary conditions are imposed on the lateral and top boundaries, whereas the bottom interface is nonabsorbing to allow light coupling and detection. Accordingly, the model comprises a 1 mm thick glass layer followed by a sample of sufficient thickness to approximate a semi-infinite scattering medium, satisfying $z \gg 1/\mu_s$. For illumination, an annular source is modeled at $z = 0$ μm, with inner and outer radii of 0.8 mm and 1.2 mm, respectively. To realize complementary illumination directions, the annular ring source is divided into four equal angular quadrants that are activated sequentially by selectively defining the corresponding angular range of the ring. For each simulation, 5×10^7 photon packets are launched with initial propagation directions sampled uniformly within a conical angular distribution spanning polar angles from 0 to $\pi/4$ relative to the local surface normal. As shown in Figure 2.2 (d), the detected photon counts per ring-source segment ranged from 247,996 to 248,952. This variation may be caused by two things: anisotropic scattering in the medium or nonuniform source irradiation from beam divergence. The latter arises because the software models extended sources as a nonuniform grid of pencil-beam arrays. Finally, photons crossing the detection plane at $z = 0$ are recorded when their positions fall within the specified detection region and their propagation directions lie within the objec-

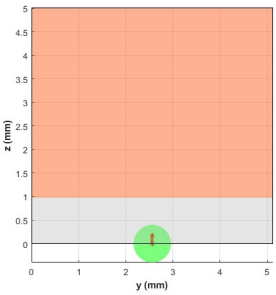
tive's acceptance cone. Specifically, the objective is assigned a numerical aperture of NA $= 0.6$, consistent with the experimental sqOBM configuration, with the acceptance half-angle defined as $\theta_{\text{acc}} = \arcsin\left(\frac{\text{NA}}{n}\right)$, where n denotes the refractive index of the immersion medium.



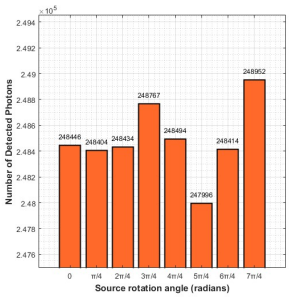
(a) Three-dimensional view



(b) XY plane



(c) XZ plane



(d)

Figure 2.2: Simulation-domain preview of the qOBM model: (a) three-dimensional view; (b) XY plane; (c) XZ plane. Legend: orange = biological tissue, gray = glass layer, pink = annular ring source, green = detector, black = no material defined. The corresponding region on the opposite side of the optical axis produces the symmetry-related contribution required by the DPC illumination geometry and therefore does not provide an independent angular distribution. (d) Illumination anisotropy characterized by the detected photon count for each of the eight segments of the annular source.

2.2 Radiative Transfer Equation (RTE)

As discussed later, DPC images are deconvolved using the corresponding optical transfer function (OTF), which is numerically calculated through a proposed microscope simulation model. For an efficient simulation of the sqOBM, it is both practical and computationally advantageous to avoid explicitly modeling a moving illumination source, as this would require recalculating the optical response of the simulated medium for each source position. This computational challenge can be avoided by exploiting the time and space reversibility principle of photon transport in a diffusive medium. This principle states that the mean optical path traversed by launched photons from the source to the detector is identical to the mean optical path that would be taken if the positions of the source and detector were interchanged [19, 20]. For each scanning beam position, it is expected that only one of the four quadrants of the 4QD will experience peak irradiance. This spatial asymmetric shift is quantified by the center of mass of the backscattered light. By invoking reciprocity, this scanning configuration is equivalent to a static forward illumination scheme. In this equivalent model, a wide-field extended source—matching the angular footprint of a single 4QD quadrant—couples a bundle of partially coherent beams into the medium. The resulting signal is then collected by a camera-based microscope objective whose optical axis passes precisely through the central aperture

of the 4QD. Consequently, the number of required simulations can be reduced from 4096 to only 4, with sqOBM represented using four large-area sources.

To simulate the optical path of the backscattered photons, a validated Monte Carlo (MC) simulator based on the RTE simulates realistic models of multiple-scattering and absorption events in turbid media [21]. Four optical parameters—absorption coefficient μ_a, scattering coefficient μ_s, anisotropy factor g which typically ranges from 0.8 to 0.95 for brain tissue in the visible to near-infrared spectral range [22], and refractive index n— fully characterize the photon-transport properties of the material. The absorption coefficient describes the rate of absorption events per unit length, and its inverse represents the mean distance between consecutive absorption events. Similarly, the scattering coefficient characterizes the rate of scattering events. In biological tissues exhibiting low absorption, optical transport is predominantly governed by molecular scattering and its associated anisotropy. The reduced scattering coefficient, $\mu_s' = \mu_s(1-g)$, is frequently reported because it is experimentally measurable. The mean scattering free path is given by $1/\mu_s$. The relationship between these optical coefficients provides information about the transport regime, indicating whether photon propagation is predominantly diffusive or ballistic.

For propagation, each photon packet launched into the medium undergoes multiple scattering and absorption events, with its weight decreasing as it propagates. Each packet is initialized at a specific source position with an initial direction and weight. The propagation step size s is sampled from an exponential distribution, and the photon's coordinates are updated accordingly. Absorption is modeled by reducing the weight of the photon packet at each interaction step. New scattering directions are then sampled according to the Henyey-Greenstein phase function [23]. Photon packets are terminated via a roulette mechanism when their weight falls below a predefined threshold or when they exit the simulation domain. By inherently accounting for multiple scattering, absorption, and stochastic trajectories, this Monte Carlo approach is exceptionally well suited for modeling light transport in optically thick, highly scattering media such as human brain tissue [24].

2.3 DPC images

Figure 2.3 shows representative differential phase contrast (DPC) images acquired with our custom-built quantitative phase imaging (QPI) microscope. The images capture the full scanned field of view, including structures partially at the boundaries. The phase gradient of identical features, such as the PMMA and MF beads, remains uniform regardless of their position across the field. This gradient defines the shear direction, the direction of highest contrast. The background exhibits a baseline phase gradient, a characteristic artifact of scanning-laser imaging configurations, as seen in (a) and (b). The phase gradient of the imaged beads is oriented opposite to this background gradient. For the larger beads, the zero-gradient line is neither vertical nor centered within the structure, unlike the symmetrical profiles of smaller beads.

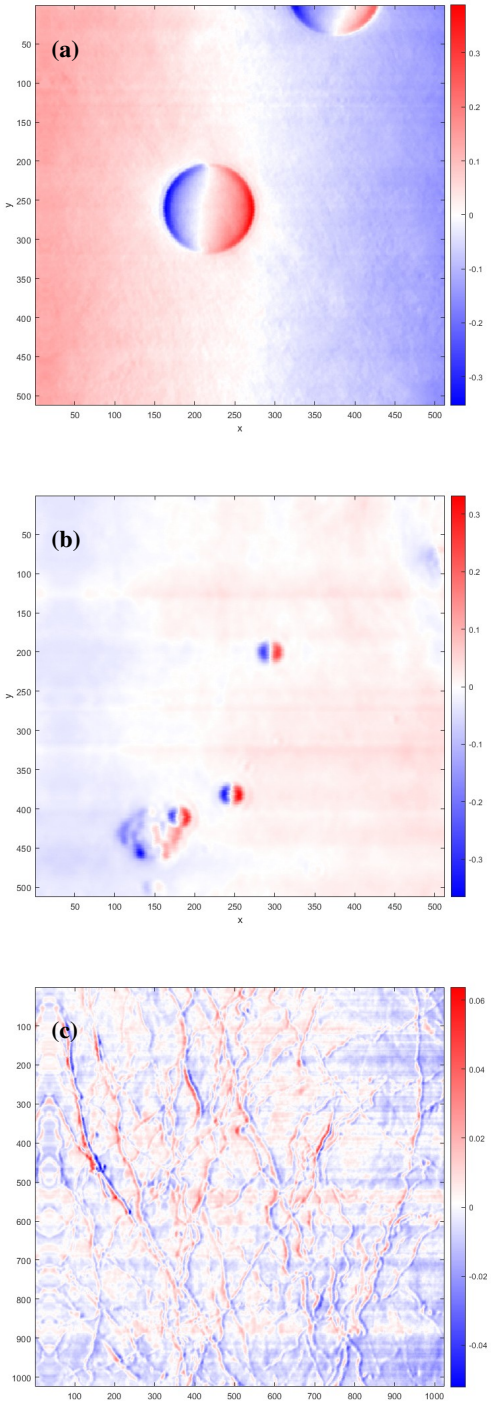


Figure 2.3: *En face* (x-y) DPC images in the x-direction acquired using the sqOBM ($\lambda = 1030$ nm, $50\times$ microscope objective, NA $= 0.8$): (a) PMMA beads placed on a glass slide and covered by a slice of pig brain white matter, (b) MF beads (11.80 µm in diameter) immersed in a scattering phantom consisting of 1% intralipid in agar, and (c) mold fungus grown on a petri dish (after background cancellation it can be seen that most of the aerial hyphal filaments (white strands radiating from the mold fungus colony) are oriented along the -45° direction). All fields of view are displayed at the same spatial scale, as indicated by the PMMA bead in (a), which has a diameter of 26 µm. In the colormap (inverted), red indicates a negative phase delay (acceleration), white indicates zero phase delay, and blue indicates positive phase delay (retardation).

3

Algorithms for phase reconstruction

Despite these high-contrast details, raw DPC data presents a directional, non-symmetric, bas-relief appearance that prevents the direct use of standard intensity-based thresholding, automated segmentation, or tracking algorithms. To overcome this limitation, a deconvolution-based algorithm must be applied. This computational step serves to decode the raw directional phase gradient data, transforming their asymmetrical profile into a symmetric, fluorescence-like intensity profile that serves as a qualitative map of the phase delay information.

Two distinct approaches to implementing the deconvolution algorithm are examined. The first employs the Hilbert transform as an image preprocessing step that converts directional contrast into a symmetric intensity profile. The second strategy relies on calculating the qOBM OTFs to perform Wiener filtering.

The final phase profile is then scaled according to the phase equation $\Delta\phi = \frac{2\pi}{\lambda}\Delta n \cdot \Delta z$, where λ is the illumination wavelength, Δn is the refractive index difference, and Δz is depth increment. The latter render the phase reconstruction presented here semi-quantitative.

3.1 Hilbert Transform

The Hilbert transform is a linear operation that has previously been employed to symmetrize real differential signals, such as differential interference contrast (DIC) images [2, 25] and photothermal images [26], and has even been used in conjunction with the transport-of-intensity equation [27]. As discussed in the previous chapter, DPC images exhibit an asymmetric, bas-relief-like intensity profile along the shear direction. The Hilbert-transform approach provides a robust means of converting this directional differential contrast into a symmetric representation [28], which can be regarded as a phase decoding function equivalent to performing a spectral integration.

– 1D Hilbert transform:

This operation provides the basis for reconstructing the phase-related signal along a selected lateral direction. In the spatial domain, it is defined as the convolution of the signal with a 1D kernel function, $h(x) = \frac{1}{\pi x}$. In the frequency domain, it is implemented by multiplying the Fourier spectrum of the signal by a transfer function, $H(u) = -i \cdot \mathrm{sgn}(u)$, where $i = \sqrt{-1}$ is the imaginary unit [2, 28]. This multiplicative factor—which is consistent with the general theoretical form of a phase OTF—introduces a phase shift of $+\pi/2$ at positive spatial frequencies, a shift of $-\pi/2$ at negative frequencies, and a zero shift to the DC component. Evaluating the Hilbert transform on the derivative of a Gaussian highlights its symmetriz-

ing behavior. The magnitude spectra confirm spectral equivalence between the derivative and its Hilbert transform, consistent with the Fourier derivative property, $\mathcal{F}\{df/dt\} = i\omega\mathcal{F}\{f\}$. Concurrently, the phase spectra demonstrate that the $\pm\pi/2$ phase shift imposed by the Hilbert transform exactly cancels the derivative's intrinsic phase. This cancellation yields a zero-phase response across the entire frequency range, mathematically validating its role as a symmetrizing operator.

Figure 3.1 shows the 1D Hilbert transform (green solid line) of the derivative (red solid line) of a 1D Gaussian signal (blue solid line) centered at $t = 0$ with a maximum amplitude of 1 and a half-width at half-maximum (HWHM) of $\sigma\sqrt{2\ln 2} \approx 1.177$ for $\sigma = 1$. The derivative (red) is antisymmetric with two opposite-polarity lobes, exhibiting a signature highly characteristic of a standard line profile extracted from an experimental DPC signal. Following this phase cancellation, the Hilbert transform successfully recovers a smooth, even-symmetric signal with a maximum amplitude of 0.966 and an HWHM of 1.202, closely resembling the original Gaussian profile except for shallow negative dips in the vicinity of its rising and falling edges.

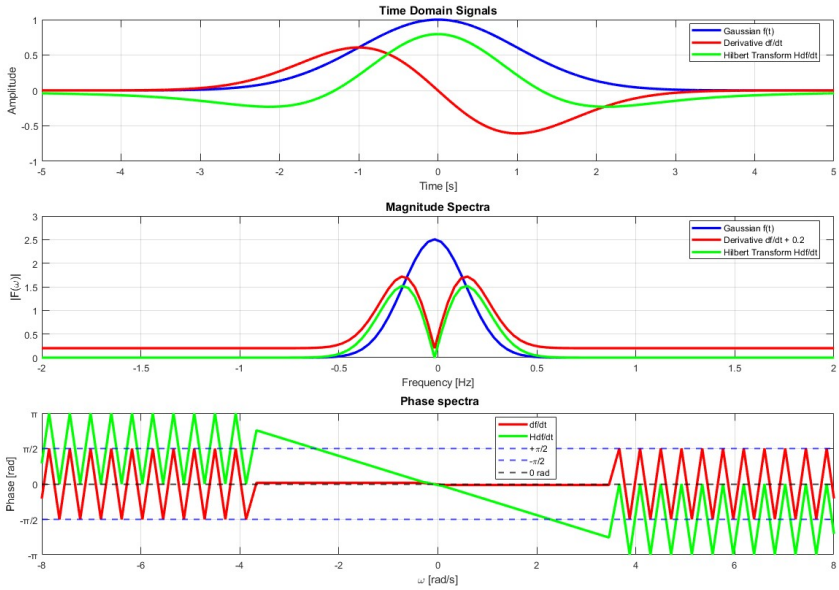


Figure 3.1: Hilbert transform applied on the derivative of a Gaussian.

– 2D Hilbert transform:

1D Hilbert transform can be applied to DPC intensity image line by line after alignment to the direction perpendicular to the shear direction rotated to match the horizontal in-plane direction of the DPC image. However, line by line, the 1D Hilbert transform suffers from degradation caused by pixel-by-pixel errors and high-frequency information suppression. A similar observation was reported when line-by-line integration is performed on DIC images [2]. Therefore, a 2D Hilbert rotatable kernel is advantageous because it can not only incorporate the two lateral spatial frequencies at the same time, allowing a one-step deconvolution, but also a directional reconstruction sensitivity is easily achieved, elevating the need for rotating the DPC images. The 2D kernel is then rotated to match the direction of the highest contrast and then upsampled to match the size of the DPC image, and the real and imaginary parts of its Fourier transform are switched after element multiplication with the kernel before taking the Fourier inverse to get the transformed image.

$H(u,v) =$ -i sgn$(\cos(\beta)\cdot u - \sin(\beta)\cdot v)$. A further extension of the 2D hilbert transform is the modified iterative Hilbert transform (MIHT), which introduces an iterative and regularized reconstruction procedure [2]. It starts with the native Hilbert transform that introduces only a constant phase shift and gradually suppresses higher frequencies with an increased number of iterations, reducing the amplification of high-frequency noise. This makes the method suitable for recovering phase-related information from noisy differential measurements while retaining the directional information encoded by the DPC illumination geometry. Additionally, setting $I_{\mathrm{DPCx}} - I_{\mathrm{DPCy}}$ as the algorithm input orients the shear direction along the diagonal, which is parallel to the $+45^{\circ}$ clockwise rotation of the x-axis, generating a directional shadow (elongation of the contrast below the background floor contrast) along the same direction. The MIHT filter is defined as:

$$\begin{cases} H_n(u,v) = -\mathrm{i}\,\mathrm{sgn}(u-v)\cdot W_n(u,v), \\ W_n(u,v) = \dfrac{\alpha}{\alpha + n(|u-v| + |u+v|)} \end{cases} \tag{3}$$

The MIHT Phase Reconstruction algorithm reconstructs a phase image from a given input image through an iterative refinement process. It begins by initializing the phase estimate to zero and setting the residual equal to the input image. Over a fixed number of iterations, the algorithm constructs the frequency-domain weighting function $W_n(u,v)$ defined above and uses it to form the Hilbert-type filter $H_n(u,v)$. The residual is transformed into the frequency domain, multiplied by this filter, and transformed back to produce a real-valued update. This update is added to the accumulated phase estimate, and its derivative is subtracted from the residual to progressively reduce the reconstruction error. After all iterations, the final accumulated phase image is returned as the output.

3.2 Quantitative Oblique Back-Illumination Microscopy (qOBM)

The downside of the Hilbert transform is that it employs a predefined kernel irrespective of the tissue under inspection or the optical configuration used for imaging. To render the reconstruction sample- and setup-specific, the kernel must be modeled accordingly. This can be achieved by computing the OTF of the qOBM microscope, which incorporates the asymmetric illumination geometry, the objective pupil function, the optical field at the image plane, and the angular distribution of the source convolved with the optical response of the tissue. The qOBM microscope uses multiple scattering to form a virtual source in the tissue slightly offset from its optical axis defined by the microscope objective, allowing a transmission scheme that illuminates the tissue from within. The qOBM-DPC imaging system allows a linear description of the optical microscope, where the measured DPC images result from convolution with the microscope's OTF (the Fourier transform of the point spread function). For an illumination direction k, this relationship in the Fourier domain is

$$\tilde{I}^{k}_{\mathrm{DPC}}(\mathbf{q}) = C^{k}_{\mathrm{DPC}}(\mathbf{q})\,\tilde{\Phi}(\mathbf{q}), \tag{4}$$

where $\tilde{I}^{k}_{\mathrm{DPC}}$ is the Fourier transform of the generated DPC image for asymmetric epi-illumination along direction k, $\tilde{\Phi}$ is the Fourier transform of the object phase distribution, and C^{k}_{DPC} is the corresponding OTF, defined as the Fourier transform of the microscope's point spread function, which can be numerically calculated according to [10] as:

$$C_{\mathrm{DPC}}(\mathbf{q}) = -i\int \left[S\left(\mathbf{u}+\frac{\mathbf{q}}{2}\right) - S\left(\mathbf{u}-\frac{\mathbf{q}}{2}\right)\right] P\left(\mathbf{u}+\frac{\mathbf{q}}{2}\right) P^{*}\left(\mathbf{u}-\frac{\mathbf{q}}{2}\right) d^{2}\mathbf{u}, \tag{5}$$

with $\mathbf{u}$ is the lateral spatial frequency coordinates at the sample plane and the integration variable over illumination angles normalized by wavelength, to sum contributions from all illumination angles, making the transfer function partially coherent. $S(\mathbf{u})$ the effective angular source distribution upsampled to match $\tilde{I}_{\mathrm{DPC}}$ Nyquist frequency, $P(\mathbf{u})$ the pupil function defined as a circular mask of radius equal to the numerical aperture of the microscope objective, and $P^*(\mathbf{u})$ its complex conjugate. $\mathbf{q}$ is the lateral spatial frequency coordinates at the image plane, $\pm\frac{1}{2}\mathbf{q}$ shifts arise from the optical field forming the intensity image result from the oblique illumination process. The term in brackets, $S(\mathbf{u}+\frac{1}{2}\mathbf{q})-S(\mathbf{u}-\frac{1}{2}\mathbf{q})$, selects the odd (anti-symmetric) component of the illumination angular distribution: Given the symmetric part of illumination does not contribute to the obtained phase contrast, only asymmetric illumination can extract phase information from a weak-phase object. So the product $P(\mathbf{u}+\frac{1}{2}\mathbf{q})P^*(\mathbf{u}-\frac{1}{2}\mathbf{q})$ represents the overlap of two shifted versions of the objective pupil. This overlap encodes the spatial frequency response of the microscope, determined by the numerical aperture and the partially coherent nature of the illumination. Additionally, equation (5) reveals two key properties of qOBM: (1) phase information is recovered only under odd effective illumination, achieved by subtracting images from two symmetrically opposed oblique illuminations, and (2) absorption is rejected by this same odd-symmetry selection, even when non-negligible. It thus provides the linear relationship between the Fourier transforms of image intensity and object phase, enabling phase recovery via deconvolution.

– Simulated OTF model:

Using the qOBM simulation model from Section 2.1 and the optical coefficients of a biological tissue listed in Table 3.1, the effective angular distribution S is retrieved for all tissues. This is done by transforming the planar histogram of the detected photons' incoming angles into a hemispherical grid and applying Gaussian smoothing (Figure **??**).

Table 3.1: Optical properties of human brain tissues [22, 24].

Tissue	μ_a	μ_s	g	n
Scalp/Skull (SS)	0.019	7.8	0.89	1.37
Cerebro-Spinal Fluid (CSF)	0.004	0.009	0.89	1.37
Gray Matter (GM)	0.02	9.0	0.89	1.37
White Matter (WM)	0.08	40.9	0.84	1.37

The results of implementing Eq. (5) for the different tissues are shown in Figure 3.2. A contour inspection across all OTF passbands reveals that the OTF associated with the CSF medium appears noisier, owing to the ballistic transport regime of propagation, whereas the remaining OTFs appear very similar. This observation can be justified by the fact that these materials share the same anisotropy coefficient, which causes the sufficiently detected photons to stochastically yield similar angular histograms of the photons forwarded toward the acceptance angle of the simulated microscope objective.

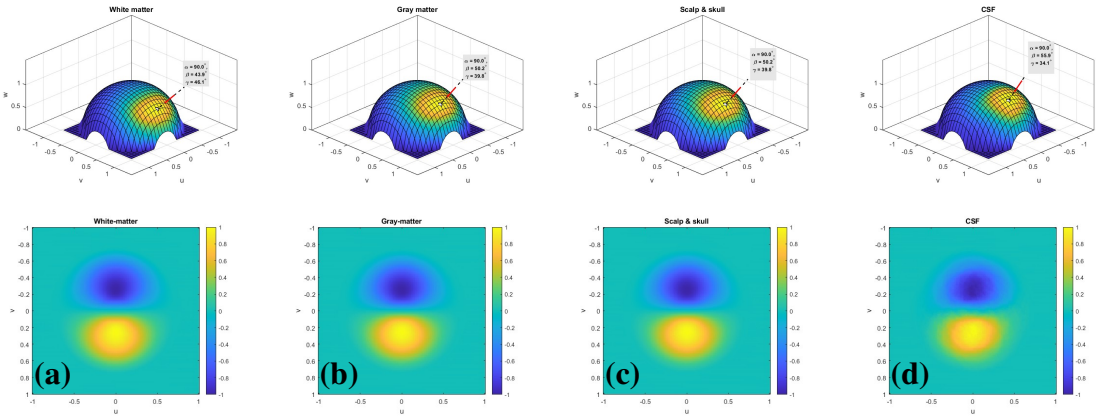


Figure 3.2: Effective source angular distributions per tissue. The height W denotes the normalized probability density. The peak weight coordinate defines the preferred photon direction (red vectors) pointing toward the grid center $(u = 0, v = 0)$. (a–d) OTFs associated with different brain tissues.

– Analytical OTF model:

A relevant observation reported by Ledwig et al. [10] is that the effective angular distribution, and consequently the calculated OTF, varies only weakly across biological tissues, owing to their highly scattering nature and near-unity anisotropy. This implies that a single OTF may suffice for phase imaging of arbitrary samples. Such a conclusion aligns with the statement by Tian et al. that "*any* complex optical transfer function provides phase contrast, by which some of the sample's phase information is translated into measurable intensity variations" [6]. To illustrate this observation, a parameter sweep systematically investigates the dependence of the optical characteristic of biological tissues and the resulting OTF. After a model simplification is employed by omitting the glass layer, direct contact between the qOBM probe and the material is simulated, while since MCXLab applies Snell's law only in the presence of a layered domain, the refractive index is dropped in this configuration, which also removes the need for an accurate refractive index mapping for every (μ_s, g) pair, and the absorption coefficient is fixed at 0.0035 for all (μ_s, g) pairs. A total number of 625 MC simulations were conducted, from which a variation of the OTF's slope at coordinates $u = v = 0$ along the $+y$-axis for a set of pairs of μ_s and g, with values ranging from 1 mm^{-1} to 40 mm^{-1} and from 0.7 to 0.9, respectively, is recorded. Figure 3.3 shows the results of this parameter sweep.

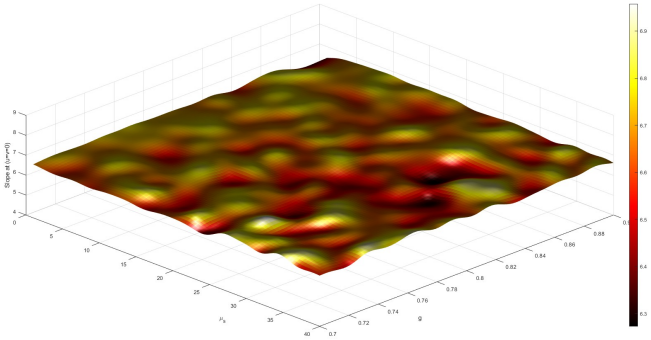


Figure 3.3: Variation of the OTF slope for different combinations of μ_s and g.

The slope is observed to exhibit only a slight variation, confined between the values of 6 and 7. Figure 3.3 is then employed as a lookup table that provides an estimate of the slope at the OTF center for every reported (g, μ_s) pair of a given biological tissue, from which the entire OTF can be reconstructed. The simulated OTF from Figure 3.2 (a) is used as support for Gaussian fitting, and a good fit was achieved with the following model:

$$OTF_{\text{model}}(u) = s_0\, u\, e^{-u^2/(2\sigma^2)}\, w(u), \tag{6}$$

Where s_0 is the slope at the center of the axis perpendicular to the asymmetry axis of the OTF, which is inversely proportional to the characteristic width of the distribution $\sigma = \frac{\sqrt{e}}{s_0}$. $w(u)$ is a function to get a good fit, defined in a window that limits the model to the relevant spatial-frequency range (given the NA = 0.6) between with $u_{\text{start}} = -0.65$ and $u_{\text{cut}} = 0.65$, The transition at the vicinity of u_{start} and u_{cut} is important to avoid discontinuities that could result from applying a sharp spatial-frequency cutoff.

$$w(u) = \begin{cases} 1, & |u| \leq u_{\text{start}}, \\ \dfrac{1}{2}\left[1 + \cos\left(\pi \dfrac{|u| - u_{\text{start}}}{u_{\text{cut}} - u_{\text{start}}}\right)\right], & u_{\text{start}} < |u| < u_{\text{cut}}, \\ 0, & |u| \geq u_{\text{cut}}, \end{cases} \tag{7}$$

For the optical properties of white-brain matter reported in Table 3.1, a value of $s_0 = 6.8$ is read from Figure 3.3 (a). The modeled 1D OTF is therefore given by $OTF_{\text{WM}}(u) = 6.8\, u\, e^{-2.6424\, u^2}\, w(u)$, as shown in Figure 3.4(c). The corresponding two-dimensional OTF is obtained by applying a full rotation of the 1D model about its center, and the resulting modeled 2D OTF is presented in Figure 3.4(b). However, this does not constitute a fully satisfactory 2D model, since the passband of the simulated OTF is non-circular, with its extent along the $+v$-axis exceeding that along the u-axis; consequently, the full rotation introduces an error of less than 0.03, which peaks in four off-axis lobes, as visible in Figure 3.4(d).

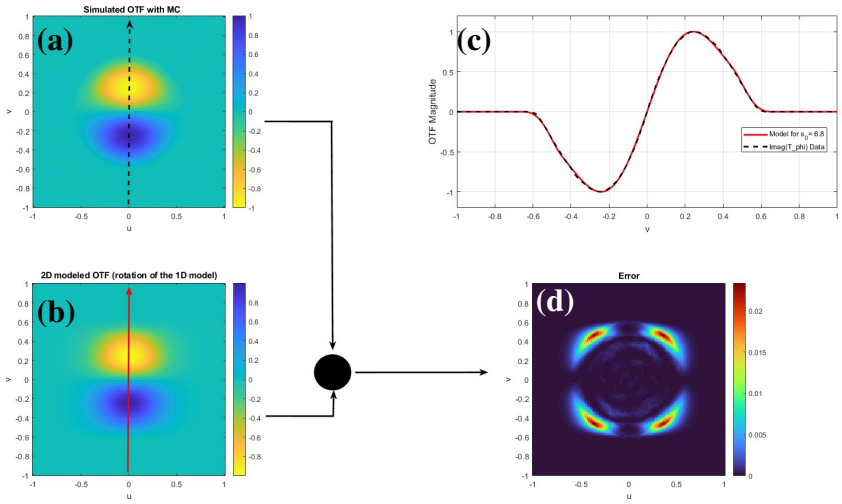


Figure 3.4: Central profile of the OTF (black dashed: MC OTF; red solid: fitted curve). Black discs indicate the absolute difference between the MC OTF and the fitted model.

– Wiener filter:

Finally, phase reconstruction from the measured DPC images requires inversion of this forward model. Recovering the phase by direct inversion of Eq. (4) exhibits a low degree of fidelity, since the DPC OTF typically contains near-zero components outside its passband. In addition, the magnitude of the OTF directly affects the estimated phase values. Consequently, direct inversion amplifies the effect of poorly transmitted spatial frequencies and introduces systematic artifacts, resulting in phase errors in the reconstructed image. The phase is therefore recovered using a Tikhonov-regularized least-squares formulation. For a set of N DPC measurements acquired with monochromatic partially coherent light sources under complementary opposite illumination directions, a least-squares problem is solved in Fourier space, minimizing the error between the DPC image and the differential image predicted by the forward model given the current reconstructed phase map, augmented by the squared modulus of the phase solution penalized by a factor α_{tik}. Rearranging yields the regularized inverse solution, and the corresponding phase distribution in real space is obtained by applying the one-step deconvolution:

$$\phi(\mathbf{r}) = \mathscr{F}^{-1}\left\{\frac{\sum_k \tilde{I}^k_{\text{DPC}} \cdot C^*_{\text{DPC}}}{\sum_k |C_{\text{DPC}}|^2 + \alpha_{tik}}\right\}. \tag{8}$$

with $\mathscr{F}^{-1}$ denoting the inverse Fourier transform, k indexing the illumination directions, and C^*_{DPC} denoting the complex conjugate of the corresponding DPC OTF. Combining measurements from multiple illumination directions increases the frequency-space coverage of the reconstruction and improves its robustness, while the Tikhonov term suppresses noise amplification in frequency regions where the OTF approaches zero. Tuning α_{tik} modifies the final retrieved phase profile, thereby postselecting the frequency content of the phase image; in the DIC framework, α_{tik} follows a 2D Gaussian distribution [2]. Here, a guess of α_{tik} as a positive constant is added to the denominator at each frequency component. Large values $\alpha_{\text{tik}} \gg 0$ cause the reconstruction to act as a high-pass filter, whereas low values $0 < \alpha_{\text{tik}} \ll 1$ cause it to act as a low-pass filter. The latter is tested by implementing the full qOBM phase reconstruction pipeline. To this end, a heterogeneous sample consisting of a three-layer structure in contact with air is considered, as shown in Figure 3.5, with different heights corresponding to SS, CSF, and WM, respectively. This setup simulates a portion of the outer layers extracted from a human brain and phase-imaged with a qOBM-DPC microscope.

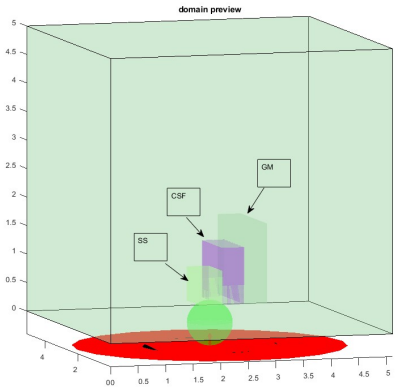


Figure 3.5: Preview of the simulation domain of three layers of the human brain in air, as imaged under the qOBM microscope (source set at a lower height relative to the detector, and the glass layer omitted for better fluence rate simulation).

Figure 3.6 shows that the three layers are clearly discernible, with better edge preservation in the lateral sections than in the axial ones, suggesting that the microscope provides good in-plane resolution. A second remark is that, despite sharing the same refractive index, the layers introduce non-zero phase delays. The latter can be justified by the fact that the optical path length, and hence the phase delay, can also be affected by differences in the optical properties of the medium, including the absorption rate and the anisotropic nature of scattering.

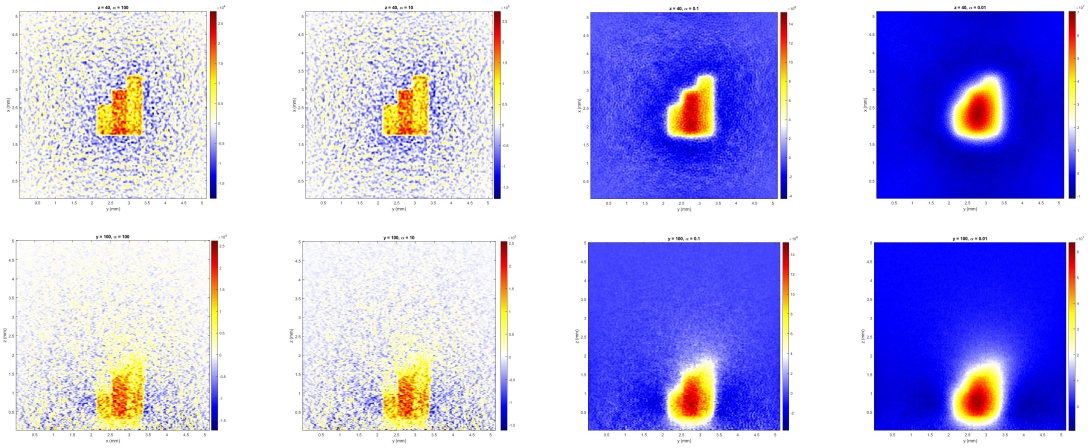


Figure 3.6: Middle and bottom rows: Reconstructed phase distributions obtained using the OTF_{WM} for xy (at $z = 0.8$ mm) and yz (at $x = 2$ mm) cross-sections, respectively. Columns from left to right correspond to the Tikhonov regularization parameter $\alpha_{tik} = 100, 10, 0.1$, and 0.01, respectively.

To evaluate the performance of the three deconvolution methodologies for 2D phase reconstruction, the samples shown in Figure 2.3 are considered and referred to as follows:

- **Method 1:** implementation based on Eq. (2.16), using the OTF shown in Figure 3.2, with $\alpha_{\mathrm{tik}} = 0.01$.
- **Method 2:** implementation based on Eq. (2.16), using a linear OTF $C_{\mathrm{DPC}} = 1 \cdot u$ for all v, with $\alpha_{\mathrm{tik}} = 0.001$.
- **Method 3:** implementation based on the Hilbert Transform (Eq. (3)), with $n = 2$ and $\alpha = 0.1$.

To obtain reasonable phase values, sample (a) was selected for phase range scaling of all other samples. By setting the background to a zero-phase delay, the maximum theoretical phase delay was found to be around 0.3691 rad (for $\lambda = 1.030$ μm, $n_{\mathrm{PMMA}} = 1.48$, $n_{\mathrm{bg}} = 1.37$, and a depth increment of $\Delta z = 0.55$ μm). Consistent with this, all three methods recovered similar phase profiles, with a quasi-constant phase value across the PMMA bead and variances relative to the predicted phase delay on the order of 0.0105 rad, 0.1175 rad, and 0.1639 rad for Methods 1, 2, and 3, respectively. The contrast-to-noise ratio (CNR) was evaluated using the average contrast at the center of the PMMA bead and the average of the largest homogeneous background region located as far as possible from the bead, yielding 59.875%, 51.085%, and 38.623% for Methods 1, 2, and 3, respectively, with Methods 1 and 2 exceeding the 50% threshold. However, upon examining samples 2 and 3, Method 2 proved superior given its lower sensitivity to background noise, as the Hilbert transform suffers from degradation due to overlap of the dark halos with the background. In the case of sample c, however, Method 3 rendered the filaments clearly visible, as they are oriented perpendicular to the shear direction, while diagonal artifacts were effectively suppressed and edges were well preserved.

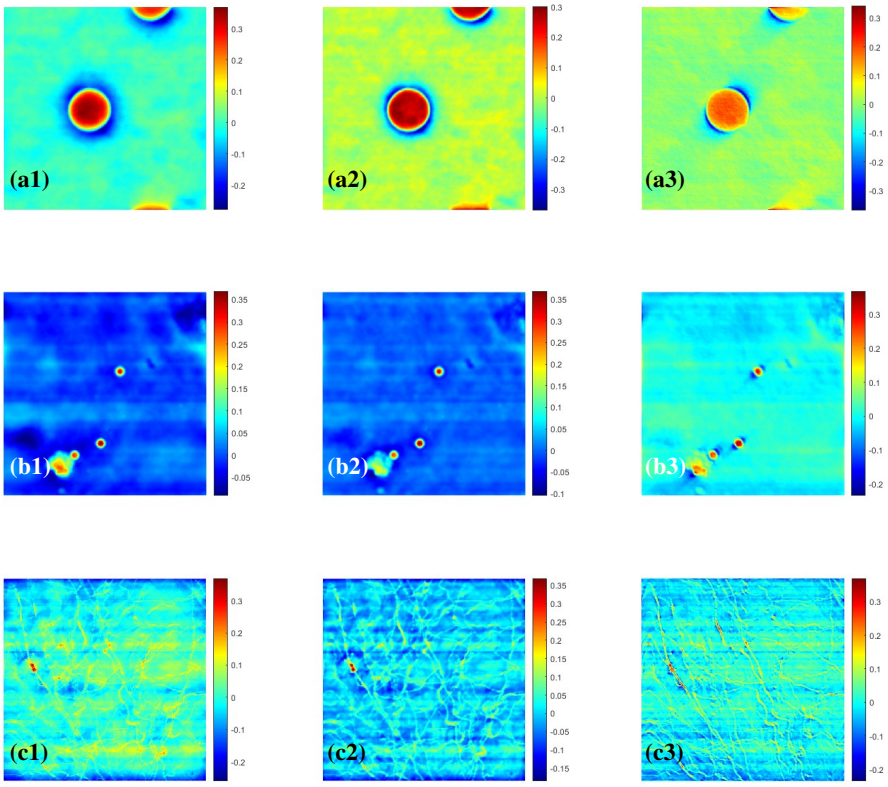


Figure 3.7: Phase reconstruction comparison. (a), (b), and (c) are the same sample as shown in Fig. 2.3. The subscripts 1, 2, and 3 denote methods 1, 2, and 3, respectively. The colormap shows phase values scaled using sample (a).

4

Conclusion and Outlook

This work was built upon three fundamental assumptions. First, most biological tissues obey the weak object approximation, which allows a linear description of the phase-intensity relation in DPC microscopy, making deconvolution the ultimate choice for phase reconstruction. Second, the single scattering event approximation extends the DPC microscope to thick, unlabeled phase objects. Third, the multislice description of semi-infinite domains enables 3D phase map retrieval.

The adopted phase imaging pipeline starts with DPC images that were experimentally acquired by measuring the center of mass of the backscattered light in a scanning optical microscope with opposite asymmetric oblique illumination, encoding the phase gradient caused by lateral variations of the refractive index at the focal plane into the output contrast image. Three samples: PMMA beads on white brain matter extracted from a young pig, mold fungi developed in a petri dish, and MF beads in an agar scattering phantom. Following optical transfer function characterization, three different 2D optical transfer functions were compared to implement the deconvolution algorithm. The optical transfer function was either proposed according to the Hilbert transform, which directly filters the DPC images to decode the phase profile, or simulated using qOBM-DPC theory. The simulated OTF is obtained by evaluating the overlap integral of the complex pupil function with its shifted conjugate, weighted by the antisymmetric component of the source angular distribution in the pupil plane. Ending by phase retrieval via deconvolution, which uses the Wiener filter formation with Tikhonov regularization to combine DPC images taken along complementary asymmetric illumination patterns with their corresponding optical transfer functions, giving directional phase sensitivity for better morphology reconstruction

Using the principle of reciprocity, a simulation model was proposed as an equivalent model of the sqOBM microscope. The open-source Monte Carlo simulator (MCXLab) models the random walk of light particles in diffusive media. The model was tested on different tissues and confirms that the forward model description of the sqOBM microscope in reflection mode, demonstrated for small sources on the order of an optical fiber cross section, is also valid for large sources on the order of a few millimeters.

For a rigorous and complete phase reconstruction, 3D optical transfer function models and regularization approaches such as generalized cross validation should be used to force the deconvolution calculation to always converge to exact values of the refractive index [29]. More accurate 2D deconvolution has been proposed by using the exact shape of the source during the OTF calculation [30], a complete treatment is proposed in the framework of 3D deconvolution for 3D imaging [10], and more robust neural networks are used to mitigate artifacts originating from the temporal and spatial coherence of the illumination, achieving fast and live 3D unlabeled imaging [31, 32].

Bibliography

[1] F. Zernike, "Phase contrast, a new method for the microscopic observation of transparent objects part ii," *Physica*, vol. 9, no. 10, pp. 974–986, 1942.

[2] B. Heise, A. Sonnleitner, and E. P. Klement, "DIC image reconstruction on large cell scans," *Microscopy Research and Technique*, vol. 66, no. 6, pp. 312–320, 2005.

[3] M. W. Davidson, D. B. Murphy, J. Hinsch, and K. R. Spring, "Comparison of phase contrast and dic microscopy." https://micro.magnet.fsu.edu/primer/techniques/dic/dicphasecomparison.html, 2015. Molecular Expressions Microscopy Primer, Florida State University. Accessed: 2026-06-27.

[4] J. Haus, "Darkfield microscopy with highpower led illumination: Optical design benefits from replacement of halogen lamps," *Optik & Photonik*, vol. 10, no. 1, pp. 29–31, 2015.

[5] T. Cacace, V. Bianco, and P. Ferraro, "Quantitative phase imaging trends in biomedical applications," *Optics and Lasers in Engineering*, vol. 135, p. 106188, 2020.

[6] L. Tian and L. Waller, "Quantitative differential phase contrast imaging in an led array microscope," *Optics Express*, vol. 23, no. 9, pp. 11394–11403, 2015.

[7] M. Chen, L. Tian, and L. Waller, "3d differential phase contrast microscopy," *Biomedical optics express*, vol. 7, no. 10, pp. 3940–3950, 2016.

[8] Y. Fan, J. Sun, Y. Shu, Z. Zhang, Q. Chen, and C. Zuo, "Accurate quantitative phase imaging by differential phase contrast with partially coherent illumination: beyond weak object approximation," *Photonics Research*, vol. 11, no. 3, pp. 442–455, 2023.

[9] C. Zuo, J. Li, J. Sun, Y. Fan, J. Zhang, L. Lu, R. Zhang, B. Wang, L. Huang, and Q. Chen, "Transport of intensity equation: a tutorial," *Optics and Lasers in Engineering*, vol. 135, p. 106187, 2020.

[10] P. Ledwig and F. E. Robles, "Quantitative 3d refractive index tomography of opaque samples in epi-mode," *Optica*, vol. 8, no. 1, pp. 6–14, 2020.

[11] P. Ledwig and F. E. Robles, "Epi-mode tomographic quantitative phase imaging in thick scattering samples," *Biomed. Opt. Express*, vol. 10, pp. 3605–3621, Jul 2019.

[12] Z. Guang, P. Ledwig, P. C. Costa, C. E. Serafini, and F. E. Robles, "Optimization of a flexible fiber-optic probe for epi-mode quantitative phase imaging," *Opt. Express*, vol. 30, pp. 17713–17729, May 2022.

[13] T. M. Abraham, P. C. Costa, C. E. Serafini, Z. Guang, Z. Zhang, S. Neill, J. J. Olson, R. Levenson, and F. E. Robles, "Label-and slide-free tissue histology using 3d epi-mode quantitative phase imaging and virtual hematoxylin and eosin staining," *Optica*, vol. 10, no. 12, pp. 1605–1618, 2023.

[14] S. Bharadwaj, Z. Guang, Z. Li, P. C. Costa, R. Chen, and F. Robles, "Real-time, in-vivo, volumetric skin imaging using quantitative oblique back-illumination microscopy," pp. TM3B–5, 2024.

[15] Z. Guang, P. C. Costa, J. Lerner, and F. E. Robles, "Toward real-time in-vivo diagnosis of brain tumors using quantitative oblique back-illumination microscopy," in *Optics and the Brain*, pp. BTh2B–3, Optica Publishing Group, 2023.

[16] S. Vyas, A.-C. Li, Y.-H. Lin, J. A. Yeh, and Y. Luo, "Isotropic quantitative differential phase contrast imaging techniques: a review," *Journal of Physics D: Applied Physics*, vol. 55, no. 18, p. 183001, 2022.

[17] L. Tian, J. Wang, and L. Waller, "3d differential phase-contrast microscopy with computational illumination using an led array," *Optics letters*, vol. 39, no. 5, pp. 1326–1329, 2014.

[18] Hamamatsu Photonics, "S11142-10: Si photodiode for electron beam detection." https://www.hamamatsu.com/jp/en/product/optical-sensors/electron_ion-sensor/photodiode-for-electron-beam-detection/S11142-10.html, n.d. Accessed April 27, 2026.

[19] J. Mertz, A. Gasecka, A. Daradich, I. Davison, and D. Coté, "Phase-gradient contrast in thick tissue with a scanning microscope," *Biomedical optics express*, vol. 5, no. 2, pp. 407–416, 2014.

[20] T. N. Ford, K. K. Chu, and J. Mertz, "Phase-gradient microscopy in thick tissue with oblique back-illumination," *Nature Methods*, vol. 9, no. 12, pp. 1195–1197, 2012.

[21] Q. Fang and D. A. Boas, "Monte carlo simulation of photon migration in 3d turbid media accelerated by graphics processing units," *Optics Express*, vol. 17, no. 22, pp. 20178–20190, 2009.

[22] A. Yaroslavsky, P. C. Schulze, I. V. Yaroslavsky, R. Schober, F. Ulrich, and H. Schwarzmaier, "Optical properties of selected native and coagulated human brain tissues in vitro in the visible and near infrared spectral range," *Physics in Medicine & Biology*, vol. 47, no. 12, pp. 2059–2073, 2002.

[23] T. Binzoni, T. S. Leung, A. H. Gandjbakhche, D. Ruefenacht, and D. Delpy, "The use of the henyey–greenstein phase function in monte carlo simulations in biomedical optics," *Physics in Medicine & Biology*, vol. 51, no. 17, pp. N313–N322, 2006.

[24] A. P. Tran, S. Yan, and Q. Fang, "Improving model-based functional near-infrared spectroscopy analysis using mesh-based anatomical and light-transport models," *Neurophotonics*, vol. 7, no. 1, pp. 015008–015008, 2020.

[25] B. Obara, M. A. Roberts, J. P. Armitage, and V. Grau, "Bacterial cell identification in differential interference contrast microscopy images," *BMC bioinformatics*, vol. 14, no. 1, p. 134, 2013.

[26] M. Li, S. Xiao, H. Ni, G. Ding, Y. Yuan, C. Marar, J. Mertz, and J.-X. Cheng, "Ultrasensitive in vivo infrared spectroscopic imaging via oblique photothermal microscopy," *Nature Communications*, vol. 16, no. 1, p. 6065, 2025.

[27] W.-S. Li, C.-W. Chen, K.-F. Lin, H.-R. Chen, C.-Y. Tsai, C.-H. Chen, and W.-F. Hsieh, "Phase retrieval by using the transport-of-intensity equation with hilbert transform," *Optics letters*, vol. 41, no. 7, pp. 1616–1619, 2016.

[28] M. Arnison, C. Cogswell, N. Smith, P. Fekete, and K. Larkin, "Using the hilbert transform for 3d visualization of differential interference contrast microscope images," *Journal of microscopy*, vol. 199, no. 1, pp. 79–84, 2000.

[29] M. Bertero, P. Boccacci, and C. De Mol, *Introduction to inverse problems in imaging*. CRC press, 2021.

[30] C. Yu, C.-E. Lin, S. Vyas, H.-P. Chiu, and Y. Luo, "Radial-balanced phase transfer functions for accurate retrieval in quantitative differential phase contrast microscopy," *Optics Express*, vol. 33, no. 14, pp. 30529–30542, 2025.

[31] Z. Li, P. Casteleiro Costa, Z. Guang, C. E. Serafini, and F. E. Robles, "Gan-based quantitative oblique back-illumination microscopy enables computationally efficient epi-mode refractive index tomography," *Biomedical Optics Express*, vol. 15, no. 8, pp. 4764–4774, 2024.

[32] P. Casteleiro Costa, S. Bharadwaj, Z. Li, N. Kaza, M. Lopez-Esteva, A. D. Lien, B. Haider, and F. E. Robles, "Single capture quantitative oblique back-illumination microscopy," *npj Imaging*, vol. 4, no. 1, p. 13, 2026.